



Acharya Institute of Technology
Department of Computer Science and Engineering
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Project Group No:

18

Project Phase I [BCS685] Synopsis on

**AI-Enabled Digital Gram Panchayat Services
Portal for Smart Rural Governance**

Submitted by

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Abstract

The **AI-Enabled Digital Gram Panchayat Services Portal for Smart Rural Governance** is a web-based system designed to improve the accessibility and efficiency of rural administrative services. In traditional Gram Panchayat systems, villagers often face delays in accessing services such as certificates, complaint registration, bill payments, and information about government schemes due to manual processes and lack of centralized management.

This project aims to create a digital platform where villagers can easily submit service requests, track application status, and receive updates online. The portal also includes AI-based features such as chatbot assistance, complaint classification, and voice input support to improve user interaction and reduce manual effort.

The system is developed using the MERN Stack, which includes MongoDB for database management, Express.js and Node.js for backend processing, and React for frontend development. AI functionalities are implemented using Python and Scikit-learn.

The proposed system helps reduce paperwork, improve transparency, minimize processing delays, and provide better communication between villagers and Panchayat authorities. It supports digital transformation in rural governance by offering a smarter and more user-friendly service platform.

1. Introduction

Rural governance in India is primarily managed through Gram Panchayats, which serve as the fundamental units of local self-government. These institutions are responsible for delivering essential public services such as issuing certificates, collecting taxes, addressing public grievances, and implementing government welfare schemes. However, most Gram Panchayat operations are still carried out manually, which leads to delays, inefficiencies, and lack of transparency. Villagers often need to visit offices multiple times to complete simple tasks, resulting in wasted time and efforts.

With the rapid advancement of digital technologies, there is a growing need to modernize rural governance systems and make services more accessible and efficient. The introduction of e-governance solutions has the potential to transform traditional administrative processes by providing online platforms for service delivery. In this context, the AI-Enabled Digital Gram Panchayat Services Portal is proposed as a smart solution that allows villagers to access services through a web-based application. This system enables users to apply for certificates, pay bills, submit complaints, and track their requests anytime and from anywhere, reducing dependency on physical visits. The proposed system is developed using modern web technologies such as the MERN stack, which includes MongoDB, Express.js, React.js, and Node.js, ensuring a scalable and user friendly interface.

AI features such as chatbot support provide instant assistance to users, while complaint detection and voice input improve interaction and automate service handling. These intelligent features help in reducing manual intervention and speeding up the overall process. Overall, the introduction of this system addresses the limitations of traditional Gram Panchayat operations by providing a digital, efficient, and transparent platform. It not only simplifies access to services for villagers but also supports Panchayat authorities in managing requests more effectively. By leveraging both web technologies and AI, the system contributes to the development of smart rural governance and promotes the vision of digitally empowered villages.

2. Literature Survey

Table 2.1 Literature Survey

S. No	Paper title	Publication details	Name of the Authors	Findings / Technical ideas discussed in the paper	Observations /Shortfalls and solution provided in the proposed methodology
1.	E-Gram Panchayat	AIJFR, Vol 7, Issue 1, 2026	Chanchal Bedse et al.	Web/mobile Portal for services,complaints, schemes, tracking, admin panel	No AI features → Add chatbot, voice input, smart routing
2.	Digital Transformation and Local Self-Governance in India	The Social Science Review, Vol 3, Issue 5, 2025	Hafijul Islam	e-Panchayat improves records, planning, transparency, participation	Infrastructure gaps, low literacy → Add multilingual AI, easy UI, automation
3.	E-Gram Panchayat	. IEEE NKCon, 2026	Keerti Patil et al.	Web application for services, admin modules, status tracking	No AI support → Add chatthatbot,
4.	GramConnect: A Digital Platform for Gram Panchayat Services	IEEE IC3ET, 2023	Bhushan Naikwade et al.	React-Firebase portal, chatbot, complaints, records, alerts	Strong portal, weak analytics → Add AI prediction, scheme suggestions,

3. Objectives

The main objective of the proposed system is to develop an AI-enabled digital platform that improves the efficiency and accessibility of Gram Panchayat services. The system is designed to simplify rural governance by providing an online portal where villagers can access essential services without depending on manual processes.

The specific objectives of the project are as follows:

- To develop a digital Gram Panchayat portal that allows users to apply for certificates, pay bills, and submit complaints online in a simple and user-friendly manner.
- To integrate Artificial Intelligence features such as chatbot assistance, complaint detection, and voice input in order to provide smart and automated service delivery.
- To design and implement an admin dashboard for Panchayat staff to verify applications, manage complaints, and monitor overall system activities efficiently.
- To enable users to track the status of their applications and complaints, such as Pending, In Progress, Approved, and Resolved, ensuring transparency in the process.
- To reduce manual workload and processing time by automating routine tasks and improving communication between villagers and Panchayat authorities.

These objectives collectively aim to enhance service delivery, increase transparency, and support the development of smart rural governance systems.

4. Proposed Methodology

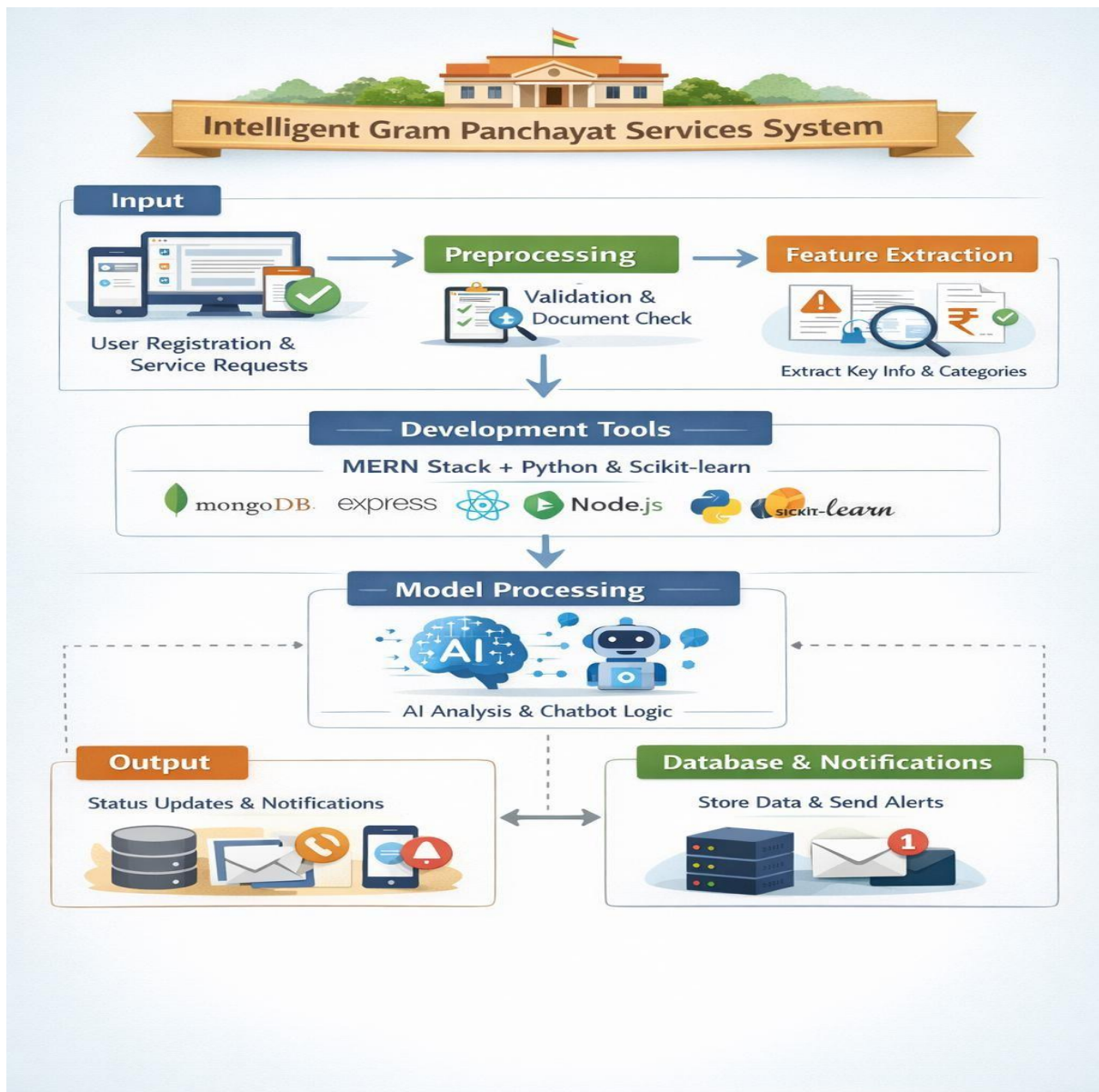


Fig 4.1 Proposed Methodology

The proposed system follows a structured approach to deliver efficient and intelligent Gram Panchayat services by integrating web technologies with Artificial Intelligence. The process begins with user interaction through a digital platform and continues through multiple stages to ensure accurate processing and effective service delivery.

In the input stage, users register or log in to the portal and submit their requests, such as applying for certificates, paying bills, or lodging complaints. This information is collected through a user-friendly web interface designed using modern frontend technologies. Once the

data is submitted, it moves to the preprocessing stage, where the system validates user details, uploaded documents, and request information. This step ensures that all inputs are complete, correct, and suitable for further processing. The system continues through multiple stages to ensure accurate processing and effective service delivery.

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After validation, the system performs feature extraction, where important details are identified from the user input. This may include recognizing the type of complaint, determining the category of service requested, or identifying eligibility for specific government schemes. These extracted features are essential for enabling intelligent processing within the system.

The implementation of the system relies on various tools and technologies, including the MERN stack (MongoDB, Express.js, React.js, Node.js) for full-stack web development. Additionally, Python and Scikit-learn are used to implement Artificial Intelligence features, while SMS and Email APIs are integrated to provide real-time notifications to users and administrators. In the model processing stage, AI components analyze the extracted data to provide automated services. The chatbot assists users by answering queries, while machine learning models classify complaints and suggest relevant schemes or services. This automation reduces manual effort and improves response time. Finally, in the output stage, the system displays the status of requests to users, stores all relevant data in the database, and sends notifications regarding updates or actions taken. This ensures transparency and keeps users informed throughout the process.

5. Requirements

5.1. Functional requirements:

- System should allow users to register/login and apply for certificates, pay bills, and submit complaints
- System should Admin dashboard should display applications, complaints, and allow status updates
- provide AI features such as chatbot, complaint detection, and voice input

5.2. Non-functional requirements:

- System should provide accurate AI predictions
- System should process requests quickly
- System should support multiple users simultaneously

5.3. Software requirements:

- React.js – Frontend framework
- Node.js – Backend runtime
- Express.js – Web framework
- MongoDB – Database • Python – Programming language
- TensorFlow/OpenCV – AI libraries

5.4. Hardware requirements:

- Processor – Intel i5 / Ryzen 5
- RAM – 8 GB
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- RAM – 8 GB

5.5. Other requirements:

- Internet connection
- SMS / Email API
- Dataset for training

6. Deliverables and Use cases

6.1. Deliverables

- AI-enabled Digital Gram Panchayat Services portal prototype that demonstrates core functionalities of the system
- Online service modules for certificate applications, bill payments, complaint registration, and government schemes access
- Web-based application developed using MERN Stack for both villagers and Panchayat staff interaction
- Admin dashboard for managing applications, verifying requests, handling complaints, and monitoring overall services
- Integration of AI features such as chatbot, complaint detection, and voice input for smart service delivery
- Notification system using SMS/Email APIs to inform users about status updates and alerts
- Well-structured project documentation including design, implementation details, and system report

6.2. Use cases

- Villagers can register and log in to the system to securely access all available services
- Users can apply for certificates online without visiting the Gram Panchayat office
- Citizens can make online bill payments quickly and efficiently through the portal
- Users can submit complaints regarding local issues and track their status in real time
- Villagers can explore and apply for suitable government schemes based on their needs
- Panchayat staff can verify applications, update statuses, and manage complaints using the admin dashboard
- AI chatbot assists users by answering queries and guiding them through different services
- System sends notifications to users and administrators regarding application progress and updates
- Multiple users can access the system simultaneously, ensuring smooth and continuous service delivery

7. Gant Chart:

Table 7.2 Gant Chart

Task	Start Date	End Date	Duration	Dependencies
Topic Selection	01-03-2026	07-03-2026	1 week	None
Literature Review	08-03-2026	28-03-2026	3 weeks	Topic Selection
Research Proposal Writing	29-03-2026	10-04-2026	2 weeks	Literature Review
Data Collection	11-04-2026	16-05-2026	5 weeks	Proposal approval
Data Analysis	29-06-2026	25-08-2026	8 weeks	Initial system developed
Thesis/report Writing	10-08-2026	25-08-2026	7 weeks	Data Analysis
Final Revisions and Editing	26-09-2026	08-10-2026	2 weeks	Report Writing
Submission	12-10-2026	15-10-2026	3 days	Final approval

8. References

- [1] Chanchal Bedse, et al., “E-Gram Panchayat,” *AIJFR*, Vol. 7, Issue 1, 2026.
- [2] Hafijul Islam, “Digital Transformation and Local Self-Governance in India: Evaluating the Impact of E-Governance on Panchayati Raj Institutions,” *The Social Science Review*, Vol. 3, Issue 5, 2025.
- [3] Keerti Patil, et al., “E-Gram Panchayat,” *IEEE NKCon*, 2023.
- [4] Bhushan Naikwade, et al., “GramConnect: A Digital Platform for Gram Panchayat Services with Chatbot Support,” *IEEE IC3ET*, 2026.